

Vigil: Trust and Risk Intelligence for Mobile Money

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1 Problem

Mobile money runs on trust. In Bangladesh, wallets like upay move everyday money for millions of people, and one scam call or one hijacked account can wipe out a user's savings along with their faith in digital payments. Fraud analysts fight this daily, but they drown in alerts. A typical system shows a bare risk score with no explanation, so every serious case costs a long manual investigation: pulling timelines, checking devices and locations, and hunting for connected wallets. Slow reviews mean stolen money keeps moving, analysts burn out, and users leave the platform. Our project tackles exactly this gap for upay: fast scoring is not enough, analysts need to understand each case and act on it with confidence.

2 Proposed idea

We proposed Vigil, an AI investigation console that answers the three questions every fraud case raises. What happened? Why is it risky? What should upay do next? Instead of another dashboard of unexplained scores, Vigil scores each transfer, explains the score in plain auditable reasons, maps it to a clear action, and writes the whole case up as a short narrative in English or Bangla. High risk never blocks money on its own. It holds the transaction, asks for step up authentication, and puts a human analyst in charge.

3 Implemented solution

The solution has three parts that work as one loop. First, a real time scoring API built with FastAPI. A transfer arrives as sender, receiver, amount, device, location, and timestamp, and the API returns a risk score between 0 and 1, a level of Low, Medium, or High, the top three reasons, a recommended action, and a written narrative. Second, an analyst console, a static web page with no build step, showing a risk sorted queue with level filters and search, a case detail view with the narrative, reasons, component scores, a mule ring panel, and a wallet timeline, plus a scoring playground and decision buttons that log Allow, Step up, or Freeze

for future retraining. Third, an offline evaluation harness that measures the system on held out data and reports ranking quality, fairness across districts and account ages, latency, and business impact. All development used synthetic data we generated ourselves plus public data, following privacy by design. No production customer data was needed or used.

4 Key features

Vigil scores transfers in about 25 milliseconds at the 95th percentile. Every score carries human readable reasons, with per transaction SHAP values backing them up. The console supports English and Bangla narratives, with a deterministic offline fallback so the demo works without internet. The transaction graph panel exposes mule rings by listing fraud wallets within two hops of a receiver. Analyst decisions feed a retraining log, and thresholds live in a config file so new on site requirements can be integrated without touching the models. Inputs are strictly validated, unauthenticated scoring traffic is isolated so it can never poison anyone's features, and the test suite covers scoring, validation, security regressions, and the language model path.

5 AI approach

Our data strategy mixes our own hardened simulator, with scam, takeover, mule, and agent patterns plus realistic noise, and PaySim, a public mobile money dataset calibrated on real logs, used only for independent validation. The model layer combines four techniques. An XGBoost classifier learns fraud shapes from tens of thousands of labeled examples. An IsolationForest learns what normal looks like from legitimate traffic alone and flags the rest, with scores calibrated to percentiles from training data. A NetworkX graph adds suspicion for receivers sitting near known fraud wallets. A weighted ensemble of 0.7, 0.2, and 0.1 blends the three into one score. A grounded language model then narrates each case from the computed facts only, under a fixed template, with sanitized inputs. On our test data the system reaches 0.998 AUC against 0.81 for hand written rules, and on PaySim it holds 0.90 while the rules collapse to a coin flip, using the same code with no tuning. We deliberately refused leakage features that would have printed perfect scores and proved nothing.

6 Intended real life impact

For analysts, Vigil turns a fifteen minute manual investigation into a two minute review, which our business simulation converts into loss prevented in BDT and analyst hours saved per thousand high risk holds. For users, faster holds on scams and takeovers mean stolen money stops moving sooner, and clear warnings protect first time and vulnerable customers before they pay a stranger. For upay, every held scam is trust kept, and trust is what grows a wallet. The API contract is stateless and ready to plug into a real stream, with weekly retraining on governed labels as the path from this prototype to a pilot. Responsible by design throughout: synthetic data only, explainable scores, fairness measured, human oversight on every consequential action.